



PAPUA NEW GUINEA NATIONAL DEPARTMENT OF HEALTH

The National Department Of Health Regulatory Bodies Nursing Council & The Medical Board Online Workforce System

Government Launch Package

Enterprise architecture, security, data governance, workflow, document management, AI, testing, deployment, and support documentation.

Generated: 07 May 2026

Readme

Government Launch Package

Project: The National Department Of Health Regulatory Bodies Nursing Council & The Medical Board Online Workforce System

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Purpose

This folder turns the platform into a formal government-grade regulatory operations package. It explains the system architecture, data governance, security controls, workflow engine, document repository, testing evidence, deployment plan, support model, and change-control process expected from an enterprise software delivery.

Package Modules

- 01 System Requirements
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Core Government Rule

Imported rows are not automatically trusted. They are staged, validated, cleansed, reviewed, approved, then promoted into live registry records.

Standards Alignment

This package aligns the platform with:

- NIST Cybersecurity Framework 2.0: <https://www.nist.gov/cyberframework>
- NIST Secure Software Development Framework SP 800-218: <https://csrc.nist.gov/pubs/sp/800/218/final>
- OWASP Application Security Verification Standard: <https://owasp.org/www-project-application-security-verification-standard/>
- OWASP Top 10: <https://owasp.org/Top10/>
- PNG Digital Government / DICT cyber security direction: <https://www.digital.gov.pg/>

Current Implementation Position

Implemented foundations:

- Nursing Council and Medical Board workspace separation.
- Role-based dashboard access.
- System Admin-only secure administration console.
- Secure password reset.
- Production-toggle MFA for System Admin and Registrar roles.
- Security audit event model for login and MFA evidence.
- Application status history and regulatory audit logs.
- OpenKM-style document repository with metadata, versions, OCR/search support, office scope, and audit events.
- Financial forecast separation by office scope.
- Data-quality audits and duplicate-review workflows.
- Staff inbox, notifications, chat, and operational access request workflow.

Launch gates still required:

- NDOH ICT hosting review and approval.
- Production email service configuration.
- Production HTTPS certificate and domain.
- Backup restore drill.
- Vulnerability scan.
- Independent penetration test.
- Registrar, finance, data-quality, and System Admin UAT sign-off.

01 System Requirements

01 System Requirements

Objective

Deliver a government-grade regulatory operations system for the National Department of Health regulatory bodies, supporting the Nursing Council and the Medical Board as separated but centrally governed workspaces.

Business Scope

The platform must support:

- Public account registration and professional self-service.
- Nursing Council application, registration, renewal, licence, workforce, and document workflows.
- Medical Board application, registration, licence, workforce, and financial workflows.
- Registrar approval and rejection decisions.
- Reviewer screening and data-quality review.
- Finance receipt tracking and financial forecast reporting.
- System Admin configuration, user approval, security, and support.
- Official document repository and records management.
- Import staging, cleansing, source tracking, and live registry promotion.
- Management reports, monthly analytics, yearly analytics, financial reports, and ministerial briefs.

Functional Requirements

ID	Requirement	Current status
FR-001	Separate Nursing Council and Medical Board dashboards, reports, and financial data.	Implemented foundation
FR-002	Enforce backend role checks, not only hidden buttons.	Implemented foundation
FR-003	Provide public-safe register search.	Implemented foundation
FR-004	Provide application intake and status tracking.	Implemented foundation
FR-005	Require document checklist, payment, review, registrar decision, and licence update.	Implemented foundation
FR-006	Store every official status change in status history.	Implemented foundation
FR-007	Store sensitive actions in audit logs.	Implemented foundation, expanding
FR-008	Provide OpenKM-style repository for scanned records, receipts, qualifications, letters, and competencies.	Implemented foundation
FR-009	Provide OCR/search and duplicate document checksum detection.	Implemented foundation
FR-010	Provide data cleansing and duplicate-review workflows.	Implemented foundation
FR-011	Provide finance forecast and office-specific financial exports.	Implemented foundation

ID	Requirement	Current status
FR-012	Require production MFA for System Admin and Registrar roles.	Implemented as production-toggle control
FR-013	Provide formal documentation, user guide, training guide, deployment plan, and support model.	Implemented in this package

Non-Functional Requirements

ID	Requirement	Target
NFR-001	Availability	Business-hours availability with documented maintenance windows
NFR-002	Performance	Dashboard pages should load within acceptable operational time after data indexing and query tuning
NFR-003	Security	Align with NIST CSF, NIST SSDF, OWASP ASVS, OWASP Top 10, and PNG Digital Government direction
NFR-004	Privacy	Users only access records within their office scope or own profile
NFR-005	Auditability	Login, MFA, import, export, approval, document, and status actions must be traceable
NFR-006	Maintainability	Changes must be modular, tested, documented, and approved before production
NFR-007	Recoverability	Backups must be restorable and tested monthly
NFR-008	Data integrity	Imported records must be staged and reviewed before becoming official live records

Acceptance Criteria

The platform is government-launch ready when:

- Production environment has HTTPS, secure cookies, restricted admin, MFA, and monitored backups.
- Registrars can approve/reject applications with status history and audit evidence.
- Finance reports show office-specific receipts and export user/timestamp.
- Repository documents are scoped, searchable, versioned, and audited.
- Data import outputs show source file, source sheet, row count, issue count, and approval status.
- UAT is signed by Nursing Council Registrar, Medical Board Registrar, Finance Officer, Data Quality Officer, and System Admin.
- External security test results are reviewed and remediated before go-live.

02 Technical Architecture

02 Technical Architecture

Enterprise Architecture View

```
flowchart LR
    Public[Public users and applicants]
    Professionals[Nurses, midwives, nurse aides, doctors, CHWs, graduands]
    Registrars[Registrars]
    Reviewers[Reviewers and data-quality officers]
    Finance[Finance users]
    Admin[System Admin]

    Web[Secure Web Platform]
    RBAC[Role-Based Access and MFA]
    Workflow[Regulatory Workflow Engine]
    Docs[OpenKM-Style Document Repository]
    Reports[Reporting and Analytics Engine]
    AI[AI Staff Assistant]
    Email[Email and Notification System]
    DB[(Primary PostgreSQL Database)]
    Backup[(Backup and Restore Server)]
    GovHost[Future Government Hosting Environment]

    Public --> Web
    Professionals --> Web
    Registrars --> Web
    Reviewers --> Web
    Finance --> Web
    Admin --> Web

    Web --> RBAC
    RBAC --> Workflow
    RBAC --> Docs
    RBAC --> Reports
    RBAC --> AI
    Workflow --> DB
    Docs --> DB
    Reports --> DB
    AI --> DB
    Workflow --> Email
    DB --> Backup
    Web --> GovHost
    DB --> GovHost
    Backup --> GovHost
```

Regulatory Workspace Separation

```
flowchart TB
    User[Authenticated user]
    Access[Backend access checks]
    Nursing[Nursing Council workspace]
    Medical[Medical Board workspace]
    General[General Registry workspace]

    NursingData[(Nursing records, receipts, documents, reports)]
    MedicalData[(Medical records, receipts, documents, reports)]
    GeneralData[(Shared public/reference data)]

    User --> Access
    Access -->|Nursing role or approved scope| Nursing
    Access -->|Medical role or approved scope| Medical
    Access -->|System Admin / approved shared scope| General
    Nursing --> NursingData
    Medical --> MedicalData
    General --> GeneralData
```

Application-To-Licence Workflow Architecture

```
sequenceDiagram
    participant Applicant
```

```

participant Portal
participant Checklist
participant Finance
participant Reviewer
participant Registrar
participant LicenceRegister
participant Reports
participant Audit

Applicant-->>Portal: Submit application
Portal-->>Checklist: Generate required documents
Checklist-->>Portal: Validate uploaded evidence
Portal-->>Finance: Verify receipt or waiver
Finance-->>Audit: Record payment verification
Portal-->>Reviewer: Send for screening
Reviewer-->>Audit: Record review action
Reviewer-->>Registrar: Forward complete file
Registrar-->>Audit: Approve or reject
Registrar-->>LicenceRegister: Issue, renew, close, or update licence
LicenceRegister-->>Reports: Refresh dashboards and statistics

```

Import Staging Architecture

```

flowchart LR
    Raw[Raw paper records and spreadsheets] --> Upload[Import upload]
    Upload --> Stage[Staging/import tables]
    Stage --> Validate[Validation and normalization]
    Validate --> Issues[Missing data and duplicate review]
    Issues --> Approve[Registrar or authorised staff approval]
    Approve --> Live[Live registry records]
    Live --> Reports[Dashboards and reports]
    Reports --> Audit[Source and audit trail]

    Raw --> Upload
    Upload --> Stage
    Stage --> Validate
    Validate -->|Clean rows| Approve
    Validate -->|Problems found| Issues
    Issues --> Approve
    Approve --> Live
    Live --> Reports
    Stage --> Audit
    Approve --> Audit

```

Deployment Architecture

```

flowchart TB
    Dev[Development environment] --> UAT[Staging / UAT environment]
    UAT --> Prod[Production environment]
    Prod --> Backup[Backup / restore environment]
    Backup --> ICT[NDOH ICT monitoring and hosting]

    Dev -->|tested release branch| UAT
    UAT -->|signed UAT release| Prod
    Prod --> Backup
    Backup -->|monthly restore drill| UAT
    Prod --> ICT
    Backup --> ICT

```

Main Components

Component	Responsibility
Public portal	Public sign-in, registration, password reset, register search, help guidance
Staff dashboards	Registrar, reviewer, finance, data-quality, and System Admin work areas

Component	Responsibility
Workflow engine	Application statuses, checklist gating, payment gating, registrar decisions, licence actions
Document repository	Official record storage, metadata, OCR/search, versions, retention, office scope, audit events
Data import layer	Source tracking, staging rows, workbook import, ATP/N-DATA alignment, cleansing queues
Reporting engine	Monthly analytics, yearly analytics, financial forecast, management brief outputs
Security layer	Role checks, MFA, secure password reset, secure cookies, session timeout, admin restriction
Notification layer	Staff inbox, chat, operational access requests, email notifications
Database	Master registry, application, licence, receipt, document, audit, import, and user data
Backup server	Scheduled database and media backups with restore testing
Government hosting	Future production hosting under approved NDOH ICT infrastructure

03 Data Governance And Dictionary

03 Data Governance and Data Dictionary

Core Data Governance Rule

Imported rows are not automatically trusted. They are staged, validated, cleansed, reviewed, approved, then promoted into live registry records.

Data Lifecycle

```
Raw paper/spreadsheet/source file
-> import batch
-> staged import rows
-> validation and normalization
-> missing-data/duplicate review
-> authorised approval
-> live registry record
-> report/dashboard/public-safe output
```

Source Evidence Requirements

Every imported or corrected record should keep:

- Source file name.
- Source sheet name.
- Source row number where available.
- Import batch.
- Import date.
- Imported by user.
- Data-quality issue notes.
- Reviewer or registrar decision where required.

Data Dictionary

Field	Plain meaning	System use	Governance rule
Registration number	The official registration identifier issued by the regulatory body.	Used to match practitioners, public register search, renewals, reports, and duplicate review.	Must be unique within the correct regulatory body unless approved as a historical duplicate exception.
Practitioner number	Internal practitioner identifier used to track one person across records.	Used for matching imports, licence rows, applications, and duplicates.	Duplicate practitioner numbers must be blocked or sent to data-quality review.
Licence number	Identifier for a specific licence or authority to practise.	Used for renewal, full registration, temporary licence, and practising licence tracking.	Do not create duplicate active licence periods for the same practitioner.
ATP	Authority To Practice record or current practising authority data.	Used for current workforce statistics, recent licence tracking, and employment alignment.	ATP data must be imported with source file, date, category, and review status.
Qualification	Education, award, certificate, or professional credential.	Supports registration eligibility, pathway decisions, and specialist recognition.	Qualification records should include institution, award title, completion year, and source evidence.

Field	Plain meaning	System use	Governance rule
Receipt	Payment evidence for application, renewal, registration, or licence.	Used by finance and registrar workflow gating.	Receipt rows must be separated by Nursing Council and Medical Board office scope.
Source sheet	Workbook tab or dataset sheet where imported row came from.	Supports traceability and management explanations.	Reports must explain whether a total came from live registry or imported historical rows.
Import batch	Group of rows imported together from one source file or workbook.	Used for rollback review, audit, and recent data-source explanation.	Each batch must record source kind, imported date, completed status, and row totals.
Province	Provincial location linked to person, workplace, facility, or institution.	Used in geographic distribution and workforce planning.	Must be normalized to approved PNG province names.
Facility	Health workplace, hospital, clinic, or employer facility.	Used for employment, workforce distribution, and employer verification.	Facility names must be standardized and classified by sector where possible.
Institution	Training school, nursing college, university, or awarding body.	Used for graduand, qualification, and education pipeline reporting.	Nursing Council local training institutions must be separated from overseas institutions and health facilities.
Employment status	Whether a practitioner is employed, unemployed, inactive, retired, deceased, or unknown.	Used for workforce planning and renewal validation.	Renewal workflows should require employment status before approval.
Employment sector	Public, church, private, NGO, overseas, or unknown workplace category.	Used in workforce-sector reporting.	Must be selected from a controlled list, not free text where avoidable.
Professional category	Nurse, midwife, nurse aide, doctor, CHW, allied health, graduand, specialist, etc.	Used for dashboards, role alignment, pathways, and counts.	Must remain inside correct regulatory body scope.
Application status	Current stage of a submitted application.	Drives review queues and registrar action.	Every official status change must write status history.
Document type	Receipt, qualification, competency, transcript, ID, police clearance, medical report, letter, etc.	Drives checklists and repository metadata.	Required documents must be verified, rejected, waived, or marked pending.
Office scope	General, Nursing Council, or Medical Board.	Separates records, receipts, documents, reports, and access.	Cross-office access must be denied unless explicitly authorised.

Trusted Versus Untrusted Data

Data type	Trust level	Required action
Raw paper records	Untrusted until captured and verified	Scan, index, extract, validate, and attach to source record
Spreadsheet imports	Untrusted until reviewed	Stage, normalize, run duplicate/missing-data checks, then approve
Live registry records	Operational source of truth	Protect with role checks, audit changes, and reconcile regularly
Public register output	Public-safe view only	Expose only name, registration number, category, status, and expiry where policy allows
Management reports	Decision-support output	Show source date, live count date, and limitations

Data Quality Controls

Minimum checks before promotion to live registry:

- Required identity fields exist.
- Registration/practitioner/licence numbers are valid or flagged.
- Duplicate name plus registration/practitioner/licence number reviewed.
- Date fields are realistic and not future-date errors such as accidental 2050 dates.
- Province, institution, facility, sector, and category are normalized.
- Receipt rows have amount, date, reference, source, and office scope.
- Qualification rows have institution, award, and completion year where available.
- Reviewer notes explain unresolved exceptions.

Production Data Preparation Command

Use the production data preparation command before handover and after major imports:

```
.\.venv\Scripts\python.exe manage.py prepare_production_data --write-report
```

This runs in dry-run mode by default. It reports what would be normalized without changing the database.

To apply safe non-destructive normalization:

```
.\.venv\Scripts\python.exe manage.py prepare_production_data --apply --skip-audit --write-report
```

The command safely normalizes:

- Leading/trailing spaces.
- Repeated internal spaces.
- Known placeholder values such as N/A, UNKNOWN, TBA, and -.
- Province naming where the mapping is obvious.
- Gender values such as M/F.
- Email casing.
- Key reference-number casing.

The command does not invent missing facts and does not overwrite suspicious date values. Those remain in missing-data, duplicate-review, or date-issue reports for staff review.

Management Reporting Rule

Reports must separate:

- Live people counts.
- Imported historical row counts.
- Receipt/payment row counts.
- Current month activity.
- Current year activity.
- Latest source file and source date.
- Cleansing limitations.

04 Security Privacy Controls Matrix

04 Security and Privacy Controls Matrix

Standards Used

Standard	Why it matters
NIST Cybersecurity Framework 2.0	Provides governance, identify, protect, detect, respond, and recover structure.
NIST Secure Software Development Framework SP 800-218	Provides secure development and release practices.
OWASP ASVS	Provides application security verification controls.
OWASP Top 10	Covers common web application risks.
PNG Digital Government / DICT direction	Aligns hosting, cyber security, and public-sector digital operations with national direction.

Control Matrix

Control area	Government-grade requirement	Platform status	Launch action
Governance	Security and privacy ownership must be assigned.	Documentation package added.	Assign NDOH ICT owner, System Admin owner, and registrar data owners.
MFA	System Admin and Registrar roles must use MFA.	Production-toggle email MFA implemented through <code>REQUIRE_STAFF_MFA=true</code> .	Configure production email and enable setting before go-live.
Role-based access	Access must be enforced in backend checks.	Implemented foundation through central role checks and scoped views.	Continue adding tests for every new workflow.
Admin access	Backend console must be restricted.	System Admin-only access is enforced.	Verify production superuser list monthly.
Password reset	Password reset must be secure and non-disclosing.	Implemented with reset tokens and neutral missing-email behavior.	Configure production email and monitor reset events.
Session security	Sessions must expire and avoid cached private pages.	15-minute idle timeout and no-cache headers exist.	Confirm production timeout policy with NDOH ICT.
HTTPS	Production must use HTTPS only.	Security settings support <code>USE_HTTPS=true</code> .	Install certificate, enable secure cookies, HSTS, and redirect.
Secure cookies	Cookies must be secure and HTTP-only where appropriate.	Settings support secure cookies and HTTP-only session cookie.	Enable <code>USE_HTTPS=true</code> , <code>SESSION_COOKIE_SECURE=true</code> , <code>CSRF_COOKIE_SECURE=true</code> .
Audit logs	Sensitive actions must be traceable.	Workflow, document, and security audit foundations exist.	Extend audit coverage to every export/import/data-edit endpoint.
Document privacy	Documents must be scoped and audited.	Office-scoped repository with audit events exists.	Train staff to store official evidence only in scoped folders.
Public privacy	Public register must expose safe fields only.	Implemented foundation.	Review public-field policy with legal/regulatory owners.
Data imports	Imported data must not become trusted automatically.	Import/audit/data-quality commands exist.	Formalize staging approval workflow in operations.
Vulnerability scanning	Scan before handover.	Not an internal code-only task.	Run dependency and web vulnerability scans before production.

Control area	Government-grade requirement	Platform status	Launch action
Penetration test	Independent test before launch.	Not an internal code-only task.	Engage independent tester and remediate findings.
Backup and recovery	Backups must be tested.	Documented in this package.	Run monthly restore drill and record evidence.
Monitoring	Production errors and security events must be reviewed.	Security audit model added.	Add production log aggregation and alerting.

Production Environment Security Settings

Set these before launch:

```
DEBUG=False
USE_HTTPS=True
SESSION_COOKIE_SECURE=True
CSRF_COOKIE_SECURE=True
SECURE_SSL_REDIRECT=True
SECURE_HSTS_SECONDS=31536000
SECURE_HSTS_INCLUDE_SUBDOMAINS=True
SECURE_HSTS_PRELOAD=True
REQUIRE_STAFF_MFA=True
STAFF_MFA_TIMEOUT_SECONDS=600
ALLOWED_HOSTS=approved-domain.gov.pg
CSRF_TRUSTED_ORIGINS=https://approved-domain.gov.pg
```

MFA Operating Rule

- MFA is mandatory for System Admin and Registrar roles in production.
- Verification codes must be sent to approved official email accounts.
- Staff without email addresses must not be granted production System Admin or Registrar access.
- MFA challenge and verification events are recorded in security audit logs.

Privacy Rules

Never expose these through public search:

- Date of birth.
- Phone number.
- Email address.
- Home address.
- Passport number.
- Police clearance.
- Medical report.
- Academic transcript.
- Receipt image.
- Internal review notes.

Pre-Launch Security Checklist

- Production DEBUG=False.
- HTTPS certificate installed.
- Secure cookies enabled.

- MFA enabled for System Admin and Registrars.
- Admin console tested with System Admin only.
- Reviewer/finance/data-quality restrictions tested.
- Public register privacy tested.
- Password reset tested using production email.
- Backup restore tested.
- Vulnerability scan completed.
- Penetration test completed.
- Critical/high findings remediated.
- Security sign-off recorded.

05 Role Access Matrix

05 Role Access Matrix

Access Principles

- Users must only access what their role and office scope require.
- Nursing Council users must not access Medical Board private records unless explicitly authorised.
- Medical Board users must not access Nursing Council private records unless explicitly authorised.
- Professionals and applicants can only access their own records.
- Finance users are read-only and must view Nursing Council and Medical Board finance separately.
- Reviewers must request approval before higher-risk operational tools unlock.
- System Admin manages configuration and security, not routine registrar decisions.

Role Matrix

Role	Can access	Cannot access	Approval required
Public user	Home page, public register search, public forms, account registration	Private records, staff dashboards, documents, receipts, imports	No
Nurse / Midwife / Nurse Aide	Own profile, own applications, own receipts, own messages, public guidance	Other practitioners, staff tools, exports, records hub	No
Graduand	Own graduand dashboard, own provisional pathway information, own applications	Staff dashboards, registrar decisions, imports	No
Doctor / CHW	Own Medical Board/CHW profile and applications	Nursing Council records, staff tools, other users	No
Reviewer	Assigned dashboard review view, staff inbox, AI Staff Assistant, limited review queues	Registrar approvals, imports, exports, operational command buttons, admin console	Yes, for elevated operations
Data Quality Officer	Duplicate review, missing-data review, Records Hub corrections where allowed	Registrar decisions, routine licence issue, System Admin tools	Yes, for elevated operations
Finance Officer	Workforce Flow, Nursing Council Financial Forecast, Medical Board Financial Forecast, finance exports	CRUD registry edits, imports, registrar approvals, admin console	Yes, for elevated operations
Nursing Council Registrar	Nursing Council dashboard, applications, receipts, documents, imports, reports, approvals	Medical Board private workspace unless authorised, System Admin configuration	MFA required in production
Medical Board Registrar	Medical Board dashboard, applications, receipts, documents, imports, reports, approvals	Nursing Council private workspace unless authorised, System Admin configuration	MFA required in production
System Admin	Secure Administration Console, user management, security settings, configuration, audit review	Routine regulatory approvals unless formally delegated	MFA required in production

Admin Console Rule

The secure administration console is reserved for System Admin only.

Registrars should use their registrar dashboards for normal operations. This keeps system configuration separate from regulatory decision-making.

Operational Access Requests

Reviewer-style users request elevated access through My Profile.

Approval path:

```
Reviewer submits request
-> registrar/System Admin receives notification
-> registrar/System Admin reviews reason
-> approve or reject
-> decision is recorded
-> user access changes only if approved
```

Finance Separation

Finance users may view both finance pages, but each page must remain office-scoped:

- /dashboard/reports/financial/?office=nursing
- /dashboard/reports/financial/?office=medical

Nursing Council and Medical Board receipts must not be merged into one uncontrolled total.

Public Register Rule

Public register output is a safe view. It must not expose private identity, contact, medical, police, transcript, receipt, or internal review information.

06 Workflow Engine Spec

06 Workflow Engine Specification

Core Workflow Rule

No silent approvals. No undocumented edits. Every official decision must have status history and audit evidence.

Standard Application Workflow

```
Application submitted
-> document checklist generated
-> payment or waiver verified
-> reviewer screening
-> registrar decision
-> licence issued, renewed, closed, rejected, or held
-> public/statistical dashboard updated
-> audit trail retained
```

Required Statuses

Recommended official status vocabulary:

- Draft.
- Submitted.
- Payment pending.
- Intake screening.
- Missing information.
- Data-quality review.
- Technical review.
- Supervisor review.
- Registrar review.
- Approved.
- Rejected.
- Withdrawn.
- Licence issued.
- Active.
- Renewal due.
- Expired.
- Suspended.
- Revoked.
- Deceased.

Checklist Gating

An application should not move to registrar decision until:

- Required identity fields are complete.
- Required documents are uploaded or officially waived.
- Required receipt/payment is verified or waived.

- Duplicate checks are completed.
- Missing-data items are cleared or accepted with documented exception.
- Required competency or supervisor assessment is complete.
- Office scope is correct.

Registrar Decision Controls

Registrar approval must:

- Update application status.
- Create status history.
- Create audit log.
- Issue, renew, close, or update licence record where applicable.
- Update dashboards and reports.
- Keep source evidence linked.

Registrar rejection must:

- Record reason.
- Update application status.
- Create status history.
- Create audit log.
- Notify applicant or staff queue where appropriate.

Licence Lifecycle

```
Provisional
-> full registration
-> practising licence renewal
-> renewal due
-> expired / suspended / revoked / deceased
```

Rules:

- One practitioner should not have duplicate active licence periods for the same office/category.
- Provisional licence should close when full registration is approved.
- Temporary licence must have start and end dates.
- Deceased practitioners must not renew unless registrar formally reopens the profile.

Workflow Audit Events

Minimum events:

- Application created.
- Application submitted.
- Document uploaded.
- Document verified/rejected/waived.
- Payment verified/waived.
- Status changed.
- Data-quality issue created/resolved.

- Registrar approved.
- Registrar rejected.
- Licence created/renewed/expired/suspended/revoked.
- Practitioner marked deceased.
- Report exported.

Workflow Performance Measures

Track:

- Applications submitted this month.
- Pending intake screening.
- Pending payment verification.
- Pending registrar approval.
- Missing-information backlog.
- Duplicate-review backlog.
- Approved this month.
- Rejected this month.
- Average days from submitted to decision.
- Renewals due within 30/60/90 days.

07 Document Records Management Sop

07 Document and Records Management SOP

Purpose

The OpenKM-style repository is the official evidence and records layer for the platform. It should hold scanned forms, receipts, qualifications, letters, competency documents, source files, and other regulatory evidence in controlled office-scoped folders.

Repository Scope

Scope	Use
General Registry	Shared non-private reference documents and common records
Nursing Council	Nursing Council applications, receipts, qualifications, competencies, letters, and evidence
Medical Board	Medical Board applications, receipts, qualifications, letters, and evidence

Document Types

Store and classify:

- Scanned application forms.
- Receipts and payment evidence.
- Qualifications and transcripts.
- Identity documents.
- Letters and correspondence.
- Competency assessments.
- Employer verification records.
- Supervisor assessment records.
- Import source workbooks.
- Management reports.
- Registrar decisions and signed outputs.

Standard Upload Procedure

- Select the correct office scope.
- Select or create the correct folder.
- Upload the document.
- Add document type.
- Add title that can be understood by another staff member.
- Add metadata:
 - Registration number.
 - Practitioner number.
 - Licence number.

- Receipt number.
- Source file.
- Source sheet.
- Source row.
- Applicant name.
- Application reference.
- Link to application or practitioner record where applicable.
- Mark official record if it is part of the legal/regulatory file.
- Set retention years where policy is known.
- Confirm audit event is recorded.

Versioning Rule

Do not overwrite official evidence silently.

If a corrected file arrives:

- Add it as a new version.
- Keep the old version.
- Add notes explaining why the version changed.
- Ensure only one version is marked current.

OCR and Search

Use OCR/extracted text for scanned documents where possible.

Search by:

- Title.
- Original filename.
- Registration number.
- Practitioner number.
- Licence number.
- Receipt number.
- Source sheet.
- Extracted text.

Duplicate Document Checks

Checksum duplicate detection helps identify repeated uploads.

If duplicate files appear:

- Confirm whether they are legitimate repeated evidence.
- Link each document correctly if repeated intentionally.
- Remove or archive accidental duplicates according to records policy.

Retention

Retention periods must be confirmed with NDOH legal/records policy.

Temporary recommended placeholders:

- Application and registration evidence: retain for life of registration plus statutory retention period.
- Receipts and finance records: retain according to finance/audit rules.
- Training/institution records: retain according to education and workforce-planning policy.
- Temporary working files: archive or dispose after confirmed promotion into official record.

Audit Trail

Document audit events should show:

- Created.
- Uploaded.
- Viewed.
- Downloaded.
- Metadata updated.
- Status changed.
- Linked to record.
- OCR processed.
- Access denied.

Privacy Rule

Never place Nursing Council private documents in Medical Board folders, and never place Medical Board private documents in Nursing Council folders.

08 Testing Qa Uat Checklist

08 Testing, QA, and UAT Checklist

Testing Principle

A government-grade system is not proven only by clicking screens. It needs repeatable test evidence, role-based access tests, import tests, report tests, security tests, and signed UAT.

Automated Test Evidence

Run before handover:

```
.\.venv\Scripts\python.exe manage.py check
.\.venv\Scripts\python.exe manage.py test apps.accounts.tests apps.dashboard.tests apps.workforce.tests
apps.notifications.tests apps.documents.tests --keepdb
```

Unit Test Areas

- User registration.
- Password reset.
- MFA challenge and verification.
- Role access helpers.
- Workflow validation rules.
- Document repository permissions.
- Document audit events.
- Financial report calculations.
- Data-quality duplicate detection.

Integration Test Areas

- Login -> MFA -> dashboard.
- Application submission -> checklist -> review -> registrar decision.
- Receipt verification -> registrar review unlock.
- Document upload -> metadata -> version -> download -> audit trail.
- Import -> missing-data audit -> duplicate review -> report refresh.
- Finance export -> audit timestamp/user -> return to correct page.
- Public register search -> safe public fields only.

Role Access Tests

Test	Expected result
Nursing Registrar opens Nursing Council dashboard	Allowed
Nursing Registrar opens Medical Board private workflow	Denied unless authorised
Medical Board Registrar opens Medical Board dashboard	Allowed
Medical Board Registrar opens Nursing private workflow	Denied unless authorised
Reviewer opens operational command tools before approval	Locked

Test	Expected result
Finance user opens Workforce Flow	Allowed
Finance user opens Nursing financial forecast	Allowed
Finance user opens Medical financial forecast	Allowed
Finance user attempts CRUD registry edit	Denied
Public user opens repository document	Denied
System Admin opens secure administration console	Allowed
Registrar opens secure administration console	Denied

Import Tests

For each workbook:

- Upload sample.
- Confirm batch created.
- Confirm row count.
- Confirm source file and sheet captured.
- Confirm invalid rows flagged.
- Confirm duplicate candidates flagged.
- Confirm future-date errors flagged.
- Confirm no row is silently dropped.
- Confirm live registry totals update only after approved import/promotion process.

Report Calculation Tests

Check:

- Live people counts are separate from imported rows.
- Receipt totals are separate from practitioner counts.
- Nursing Council finance is separate from Medical Board finance.
- Monthly totals use correct date fields.
- Yearly totals exclude bad future dates.
- Exported reports show source date, generation timestamp, and exporting user.

Security Tests

- Password reset does not disclose missing email addresses.
- System Admin and Registrar MFA works when `REQUIRE_STAFF_MFA=true`.
- Secure cookies enabled in production.
- CSRF protection works for POST forms.
- Admin console restricted to System Admin.
- Public register does not expose private fields.
- Access denied attempts are audited where implemented.

UAT Sign-Off

Sign-off role	Must test
Nursing Council Registrar	Nursing workflows, reports, approvals, repository evidence, public register
Medical Board Registrar	Medical workspace separation, medical reports, approvals, finance separation
Finance Officer	Nursing and Medical financial forecasts, exports, receipt explanations
Data Quality Officer	Missing-data audit, duplicate review, source traceability, cleansing SOP
System Admin	User roles, MFA, admin console, backups, deployment checklist

UAT Decision

Use one of these outcomes:

- Approved for production.
- Approved with minor issues.
- Not approved until listed issues are fixed.

All issues must have owner, severity, target date, and closure evidence.

09 Deployment Backup Support Plan

09 Deployment, Backup, and Support Plan

Environment Model

Environment	Purpose	Data rule
Development	Build and local testing	May use sanitized sample data
Staging / UAT	Registrar and staff testing before release	Use controlled copy or sanitized production-like data
Production	Live government operations	Real data only; restricted access
Backup / Restore	Recovery testing	Encrypted backup copies; tested monthly

Release Flow

```
Development change
-> automated checks
-> developer testing
-> staging deployment
-> UAT sign-off
-> production deployment window
-> post-deployment smoke test
-> release notes filed
```

Production Deployment Requirements

- NDOH ICT hosting approval.
- Approved domain and HTTPS certificate.
- PostgreSQL production database.
- Secure media/document storage.
- SMTP email service for password reset and MFA.
- Backup location separate from production server.
- Environment variables configured securely.
- DEBUG=False.
- REQUIRE_STAFF_MFA=True.
- System Admin account confirmed.
- Test/default passwords removed.

Backup Plan

Backup these:

- PostgreSQL database.
- Media/document uploads.
- Environment configuration record.
- Static release package.
- Generated official reports where required.

Minimum schedule:

- Daily database backup.
- Daily media backup.
- Weekly full backup.
- Monthly restore drill.
- Backup retention aligned with NDOH ICT policy.

Restore Drill

Monthly:

- Select latest backup.
- Restore into backup/restore environment.
- Confirm login works.
- Confirm dashboards load.
- Confirm repository documents open.
- Confirm reports generate.
- Record restore start time, finish time, and issues.
- File restore evidence.

Monitoring

Monitor:

- Login failures.
- MFA failures.
- Admin console access.
- Import errors.
- Report/export failures.
- Background command failures.
- Database storage usage.
- Media storage usage.
- Backup success/failure.
- Application errors.

Production Smoke Test

After deployment:

- Open home page.
- Log in as System Admin using MFA.
- Log in as Nursing Registrar using MFA.
- Open Nursing Council dashboard.
- Open Medical Board dashboard.
- Open financial forecast for Nursing.
- Open financial forecast for Medical.

- Open document repository.
- Run public register search.
- Confirm /admin/ denies non-System Admin users.

Support Channels

Recommended:

- Level 1: Registrar office super users.
- Level 2: System Admin / ICT application support.
- Level 3: Developer/vendor technical support.

Incident Categories

Severity	Example	Response target
Critical	System unavailable, data exposure, login outage	Immediate response
High	Registrar cannot approve, finance exports broken, document repository inaccessible	Same business day
Medium	Report display issue, single workflow bug, import row issue	2-3 business days
Low	Text correction, minor UI improvement, non-urgent enhancement	Planned release

Disaster Recovery Objective

Initial recommended targets:

- Recovery Time Objective: within 24 hours for controlled launch.
- Recovery Point Objective: last daily backup.

These targets should be tightened after NDOH ICT confirms infrastructure and staffing.

10 Staff Training Guide

10 Staff Training Guide

Training Goal

Train staff to use the system as a controlled regulatory operations platform, not as a simple spreadsheet replacement.

Training Groups

Group	Training focus
Public/applicant support staff	Account help, password reset, public forms, applicant guidance
Registrars	Dashboards, approvals, reports, office separation, audit responsibility
Reviewers	Review queues, missing data, document checks, access-request process
Finance users	Receipt review, financial forecast, separated office finance, exports
Data-quality officers	Duplicate review, missing data, import source traceability
System Admin	Users, roles, MFA, secure console, backups, deployment checks

Core Messages For All Staff

- Do not mix Nursing Council and Medical Board data.
- Do not treat imported rows as automatically correct.
- Do not publish statistics before audit and review.
- Do not download or share confidential exports without approval.
- Use the document repository for official evidence.
- Use metadata so documents can be found later.
- Use the staff inbox/chat for auditable communication.

Registrar Training

Registrars must be able to:

- Open their correct regulatory dashboard.
- Review application status.
- Check required documents.
- Confirm payment/waiver status.
- Review duplicate/missing data warnings.
- Approve or reject with reason.
- Confirm licence/register update.
- Generate reports after data updates.
- Explain source dates and live counts.
- Use MFA in production.

Reviewer Training

Reviewers must be able to:

- Understand what they can and cannot access.
- Submit operational access request if higher access is needed.
- Review assigned records.
- Check documents and metadata.
- Flag missing data.
- Avoid registrar-only decisions.

Finance Training

Finance users must be able to:

- Open Nursing Council financial forecast.
- Open Medical Board financial forecast.
- Keep the two finance scopes separate.
- Understand manual receipts versus spreadsheet-imported receipts.
- Generate PDF/Excel/Word financial reports.
- Explain monthly/yearly totals and data-source limitations.

Data-Quality Training

Data-quality users must be able to:

- Run missing-data audits.
- Review duplicate candidates.
- Check registration number, practitioner number, licence number, names, dates, institution, facility, and province.
- Record correction notes.
- Avoid deleting records without approval and evidence.

Document Repository Training

Staff must practise:

- Uploading a scanned receipt.
- Uploading a qualification.
- Adding metadata.
- Linking to a practitioner or application.
- Searching by receipt number.
- Uploading a corrected version.
- Reading audit trail.

System Admin Training

System Admin must be able to:

- Manage accounts and role approvals.
- Enforce MFA.
- Review security audit events.
- Restrict secure console access.
- Confirm backups.
- Deploy approved releases.
- Support password reset/email configuration.
- Maintain documentation and change logs.

Training Evidence

Each training session should record:

- Date.
- Trainer.
- Attendees.
- Role group.
- Topics covered.
- Issues raised.
- Follow-up actions.
- Attendance sign-off.

11 Maintenance Sla Change Request

11 Maintenance, SLA, and Change Request Model

Maintenance Purpose

Maintenance keeps the platform stable, secure, updated, backed up, and aligned with changing regulatory operations.

Monthly Maintenance Tasks

- Review user accounts and access approvals.
- Review System Admin users.
- Review security audit events.
- Review failed login/MFA activity.
- Review backup success.
- Perform backup restore drill.
- Run missing-data audit.
- Run duplicate audit.
- Review finance date errors and suspicious receipt rows.
- Confirm reports generate correctly.
- Review document repository metadata gaps.
- Apply security patches and dependency updates.
- Update documentation where workflows changed.

Recommended SLA

Service item	Target
Critical production outage	Same-day emergency response
High-impact workflow failure	Same business day investigation
Medium issue	2-3 business days
Low issue / wording / UI adjustment	Planned maintenance release
Monthly health check	Once per month
Backup restore test	Once per month
Security review	Quarterly or after major change

Change Request Process

```
Request submitted
-> classify as bug, enhancement, report, security, data, or training
-> assess risk, affected roles, affected data, and estimated cost
-> approve or reject
-> implement in development
-> test
-> deploy to staging/UAT
-> user sign-off
-> deploy to production
-> update documentation
```

Change Request Form Fields

- Request title.
- Requested by.
- Date requested.
- Regulatory body affected.
- User roles affected.
- Description.
- Business reason.
- Urgency.
- Data affected.
- Reports affected.
- Privacy/security risk.
- Acceptance criteria.
- Approved by.
- Target release.
- Completion evidence.

Pricing Model For Future Work

Recommended commercial structure:

- Monthly maintenance retainer.
- Separate fee for new modules.
- Separate fee for major data cleansing.
- Separate fee for training.
- Separate fee for penetration testing support/remediation.
- Separate fee for deployment/migration to NDOH ICT infrastructure.

Change Categories

Category	Example	Approval required
Bug fix	Broken export, incorrect redirect, chart loading issue	System Admin / product owner
Data correction	Cleansing duplicate or correcting institution	Registrar/data owner
Report update	New ministerial table or finance breakdown	Registrar/finance owner
Security change	MFA, access restriction, audit expansion	System Admin and NDOH ICT
New workflow	New pathway, new licence type, new approval step	Registrar and project owner
Infrastructure	Server, database, backup, domain, email	NDOH ICT

Release Notes

Every production release should record:

- Version/date.
- Changes included.

- Database migrations.
- Settings changed.
- Tests run.
- Known issues.
- Rollback plan.
- Approved by.

12 Gap Register And Modular Roadmap

12 Gap Register and Modular Roadmap

Purpose

This register separates completed platform foundations from launch gates and future enterprise enhancements. This prevents the project from being oversold while still showing clear government-grade progress.

Current Foundations Completed

Area	Current position
Workspace separation	Nursing Council and Medical Board are separated through dashboard and backend scope checks.
Role-based access	Central role helpers and locked reviewer/finance flows exist.
Password reset	Secure password reset flow exists.
MFA	Production-toggle MFA added for System Admin and Registrar roles.
Security audit	Login and MFA security audit event model added.
Workflow audit	Regulatory status history and audit log foundations exist.
Document repository	Office-scoped OpenKM-style repository exists with metadata, versions, search, checksum duplicate checks, and audit events.
Financial forecast	Nursing Council and Medical Board finance pages are office-scoped.
Data quality	Missing-data and duplicate-review tools exist.
Documentation	User guide, OpenKM guide, data cleansing plan, and government launch package exist.

Launch Gate Gaps

Gap	Risk	Required action	Owner
Production hosting not yet approved by NDOH ICT	High	Confirm server, domain, HTTPS, backup, monitoring, and deployment process	NDOH ICT / System Admin
MFA not enabled in local/dev by default	Medium	Set <code>REQUIRE_STAFF_MFA=true</code> in production	System Admin
Production email not configured	High	Configure SMTP for password reset and MFA	NDOH ICT / System Admin
Penetration test not completed	High	Engage independent tester and remediate findings	Project owner / NDOH ICT
Vulnerability scan not completed	High	Run dependency and web app scans before handover	System Admin / ICT
Backup restore drill not completed	High	Restore latest backup into test environment and record evidence	NDOH ICT
Full UAT sign-off not recorded	High	Obtain sign-off from registrar, finance, data quality, System Admin	Project owner
Broader audit coverage still expanding	Medium	Add audit events to all imports, exports, and high-risk edits	Developer / System Admin

Gap	Risk	Required action	Owner
Live employment data still dependent on clean ATP/renewal capture	Medium	Continue ATP alignment and make renewal employment capture mandatory	Registrar / Data Quality

Modular Enterprise Roadmap

Module 1: Security Hardening

Status: Started.

Deliverables:

- MFA production toggle.
- Security audit events.
- HTTPS production settings.
- Admin restriction verification.
- Vulnerability scan.
- Penetration test.

Module 2: Data Governance

Status: Started.

Deliverables:

- Data dictionary.
- Import staging rule.
- Source traceability.
- Missing-data SOP.
- Duplicate-review SOP.
- Institution/facility/province normalization.

Module 3: Workflow Control

Status: Started.

Deliverables:

- Checklist gating.
- Payment gating.
- Reviewer queue.
- Registrar approval.
- Licence lifecycle.
- Status history and audit trail.

Module 4: Records Management

Status: Started.

Deliverables:

- Office-scoped repository.
- Metadata.

- Versions.
- OCR/search.
- Retention.
- Audit events.
- Link evidence to applications/practitioners.

Module 5: Reporting and Finance

Status: Started.

Deliverables:

- Monthly analytics.
- Yearly analytics.
- Financial forecast.
- Export timestamp/user tracking.
- Source-date explanations.
- Ministerial/registrar brief generation.

Module 6: Deployment and Operations

Status: Documentation ready; production action required.

Deliverables:

- Dev/staging/production environments.
- Backup/restore environment.
- Deployment guide.
- Monitoring.
- Support model.
- Change request process.

Definition of Done For Government Launch

The system should not be declared production-live until:

- All critical launch gates are closed.
- MFA is enabled for production System Admin and Registrars.
- Production email works.
- HTTPS and secure cookies are enabled.
- Backup restore drill is completed.
- Vulnerability scan and penetration test are completed.
- UAT is signed.
- Staff training is delivered.
- Production support owner is assigned.

13 Enterprise Ui Design Standard

13 Enterprise UI Design Standard

Purpose

This design standard defines the production look and usability rules for the government regulatory operations platform.

Design Direction

The interface should feel:

- Official.
- Calm.
- Clear.
- Evidence-based.
- Accessible.
- Suitable for registrars, finance users, reviewers, and public users.

It should not feel like a test site, template demo, or experimental dashboard.

Visual System

Token	Use
Navy	Primary government header/sidebar identity
Teal/green	Approved, active, primary workflow action
Gold	Warnings, important notices, official highlight
Red	Rejected, failed, expired, critical risk
White cards	Main content blocks
Light grey-blue surface	Page background and table context

The shared production stylesheet is:

```
static/css/government-enterprise.css
```

Admin console styling is:

```
static/css/ndoh-admin.css
```

Layout Rules

- Keep side navigation consistent.
- Keep top-bar account/settings/logout actions predictable.
- Use cards for grouped information.
- Use tables for records and auditable data.
- Use dashboard number cards only for high-value totals.
- Use explanatory text beside reports, finance figures, and imports.
- Avoid decorative clutter that competes with official data.

Accessibility Rules

- Text must have strong contrast.
- Buttons must be visually clear and readable.
- Do not rely on color alone to communicate status.
- Tables must have readable headings.
- Forms must have labels.
- Mobile screens must remain usable.

Workflow Clarity Rules

Every operational page should answer:

- What is this page for?
- Who is allowed to use it?
- What should the user do first?
- What happens after clicking a button?
- What data source is being shown?
- What must be reviewed before management reporting?

Status Display Rules

Use standard status language:

- Pending.
- Under Review.
- Missing Information.
- Approved.
- Rejected.
- Active.
- Expired.
- Suspended.
- Resolved.

Report Display Rules

Reports must show:

- Office scope.
- Generated date/time.
- Exporting user where applicable.
- Live count versus imported row count.
- Source file/date where applicable.
- Remaining data-quality limitations.

Production UI Gate

Before launch:

- Smoke-test public home, login, MFA, dashboard, records hub, document repository, financial forecast, public register, profile, and admin console.
- Confirm no placeholder/debug/test labels remain visible.
- Confirm Nursing Council and Medical Board screens are visually consistent but data-separated.
- Confirm mobile layout is acceptable.

14 Ai Staff Assistant And Import Cleansing

AI Staff Assistant and Import Cleansing Guide

Purpose

The platform supports a controlled AI layer for two approved purposes only:

- Staff operational questions for Admin, Registrar, Reviewer, Finance, and Data Quality users.
- Import filtering and cleansing suggestions before spreadsheet rows are trusted.

Public applicants, graduands, nurses, doctors, CHWs, and other professional users do not receive this staff AI access. They use the normal public AI Helpdesk only.

Default Mode: Local Offline Assistant

The default setting is:

```
AI_ASSISTANT_PROVIDER=local
AI_ASSISTANT_EXTERNAL_ENABLED=False
AI_IMPORT_CLEANSING_EXTERNAL_ENABLED=False
```

In this mode the system does not send data outside the server. Staff answers are generated from local rules and live scoped counts. Import cleansing uses local validation rules, aliases, fuzzy matching, missing-field checks, future-date checks, old-date checks, and identifier normalization.

Optional Mode: OpenAI GPT

OpenAI API use is not treated as a free/offline feature. It requires:

- An OpenAI API key.
- Internet access from the server.
- Approved billing for token usage.
- NDOH ICT approval before any production use.

OpenAI can be enabled only if NDOH ICT approves external API use and the environment variables are configured:

```
AI_ASSISTANT_PROVIDER=openai
AI_ASSISTANT_EXTERNAL_ENABLED=True
OPENAI_API_KEY=...
OPENAI_MODEL=...
```

Import cleansing through OpenAI remains separately locked:

```
AI_IMPORT_CLEANSING_EXTERNAL_ENABLED=True
```

This separate switch prevents external GPT use for spreadsheet rows unless ICT deliberately enables it.

Recommended Free Mode: Ollama Local GPT

For a free live chat model, the recommended first option is Ollama running locally or on an internal NDOH server. Ollama exposes a local API on port 11434, can run without OpenAI billing, and supports local chat models once those models are downloaded.

Enable only after the runtime and model are approved:

```
AI_ASSISTANT_PROVIDER=ollama
AI_ASSISTANT_OLLAMA_ENABLED=True
AI_OLLAMA_BASE_URL=http://127.0.0.1:11434
AI_OLLAMA_MODEL=llama3.2:3b
AI_IMPORT_CLEANSING_MODEL_ENABLED=False
```

Import cleansing remains separately locked. If approved, turn it on with:

```
AI_IMPORT_CLEANSING_MODEL_ENABLED=True
```

Check the model server:

```
.\.venv\Scripts\python.exe manage.py ai_model_status
```

If Ollama is not running or the model is missing, the system falls back to the safe local rule-based assistant instead of crashing.

Optional Mode: Private Offline GPT

The platform also supports a provider slot for a private GPT-style model server. This is the safer path if NDOH wants offline capability because the model can run inside the approved government network instead of sending prompts to an external API.

Enable only after ICT approves the runtime, hardware, model licence, and security controls:

```
AI_ASSISTANT_PROVIDER=local_llm
AI_ASSISTANT_LOCAL_LLM_ENABLED=True
AI_LOCAL_LLM_BASE_URL=http://127.0.0.1:11434
AI_LOCAL_LLM_MODEL=approved-local-model-name
```

The local model endpoint is expected to provide an OpenAI-compatible `/v1/chat/completions` API. This keeps the platform neutral: ICT may later approve a local runtime such as Ollama, llama.cpp server, or another internally hosted model gateway without changing staff screens.

Safety Rules

- AI never approves applications.
- AI never writes directly into live practitioner tables.
- AI never runs raw SQL.
- AI never bypasses Nursing Council and Medical Board office separation.
- AI suggestions must be reviewed by staff before being promoted into live registry records.
- Import rows remain staged, validated, reviewed, approved, then promoted.
- Sensitive information should be minimized before external API use.

Import Cleansing Preview

Use this command to preview cleansing results without changing the database:

```
.\.venv\Scripts\python.exe manage.py ai_cleansse_import_preview --file path\to\workbook.xlsx --sheet "Sheet1" --rows 25 --scope nursing
```

For CSV:

```
.\.venv\Scripts\python.exe manage.py ai_cleansse_import_preview --file path\to\data.csv --rows 25 --scope nursing
```

Optional JSON output:

```
.\.venv\Scripts\python.exe manage.py ai_cleansse_import_preview --file path\to\data.csv --rows 25 --scope nursing --output docs\reports\ai_import_preview.json
```

What The Cleanser Checks

- Blank or sentinel values such as N/A, TBA, UNKNOWN, and -.
- Registration, practitioner, licence, ATP, receipt, and reference number formatting.
- Province aliases such as NCD, SHP, ENB, and AROB.
- Unknown or fuzzy province values that require staff review.
- Gender aliases such as M, F, Male, and Female.
- Future dates such as accidental 2050 payment dates.
- Old dates before 2000 that need source verification.

- Rows without usable name fields.
- Rows without usable identifier fields.

Open Source / Offline Options

If NDOH wants a fully offline GPT-style model, the recommended future pattern is to run an internal model server inside the government network, then connect it behind the same provider interface. Examples of possible offline runtimes include Ollama or llama.cpp-style deployments, subject to ICT security review, hardware capacity, model licensing, and data governance approval.

The current implementation is intentionally provider-neutral so the platform can use:

- Local deterministic rules now.
- Free local Ollama GPT when installed and approved.
- OpenAI API if approved.
- A future internal model server without changing the staff workflow.

What Staff Must Remember

AI is an assistant, not the registrar. It can explain queues, suggest cleansing actions, and flag suspicious imported data. It must not approve applications, issue licences, alter receipt totals, or promote spreadsheet rows into live registry records without human review and audit logging.

Official OpenAI References For ICT Review

- OpenAI API pricing: <https://openai.com/api/pricing/>
- OpenAI API data controls: <https://platform.openai.com/docs/guides/your-data>
- Responses API reference: <https://platform.openai.com/docs/api-reference/responses/create>
- Structured Outputs guide: <https://platform.openai.com/docs/guides/structured-outputs>
- Ollama API reference: <https://docs.ollama.com/api>
- Ollama OpenAI-compatible local API: <https://docs.ollama.com/openai>
- llama.cpp OpenAI-compatible server: <https://www.mintlify.com/ggml-org/llama.cpp/inference/server>

15 Free Local Gpt Setup

Free Local GPT Setup Guide

Purpose

This guide explains how to run the Staff AI Assistant without paid OpenAI API usage. The platform now supports a free local model provider, with Ollama as the recommended first option.

The assistant remains restricted to staff roles and is used only for:

- Staff operational questions.
- Data-quality guidance.
- Import-cleansing suggestions before live database promotion.
- Report and workflow guidance.

It must not approve applications, issue licences, update receipt totals, or promote imported rows into live records without human review.

Recommended Free Option: Ollama

Ollama is practical for this project because it can run locally, exposes a simple chat API, and supports an OpenAI-compatible local API. This means the platform can use a live GPT-style model without sending data to a paid external API.

Suggested setup pattern:

```
ollama pull llama3.2:3b
ollama serve
```

Then configure the platform environment:

```
AI_ASSISTANT_PROVIDER=ollama
AI_ASSISTANT_OLLAMA_ENABLED=True
AI_OLLAMA_BASE_URL=http://127.0.0.1:11434
AI_OLLAMA_MODEL=llama3.2:3b
AI_IMPORT_CLEANSING_MODEL_ENABLED=False
```

Use a small model first so the laptop or server remains responsive. Larger models may answer better but require more RAM, disk space, and CPU/GPU capacity.

Health Check

After starting the model server, run:

```
.\.venv\Scripts\python.exe manage.py ai_model_status
```

Expected result:

```
Configured provider: ollama
Active mode: ollama
Ollama is reachable. Installed models:
- llama3.2:3b
```

If Ollama is not running, the command will warn staff and the platform will keep using the local rule-based fallback assistant.

Import Cleansing

Model-assisted import cleansing is deliberately separated from staff chat.

Keep it off during normal testing:

```
AI_IMPORT_CLEANSING_MODEL_ENABLED=False
```

Only turn it on after ICT approves the model and import data-handling controls:

```
AI_IMPORT_CLEANSING_MODEL_ENABLED=True
```

Preview a file without changing the database:

```
.\.venv\Scripts\python.exe manage.py ai_cleanse_import_preview --file path\to\data.xlsx --rows 25  
--scope nursing
```

Other Free / Open-Source Options

Ollama is the easiest first integration. Other options can be used later if ICT prefers them:

- llama.cpp server: useful for a lightweight internal model server with OpenAI-compatible endpoints.
- LocalAI: useful when ICT wants an internal OpenAI-compatible gateway for several model types.
- Hugging Face Transformers: powerful but heavier to package and maintain inside a production web system.

Any option must be reviewed for licensing, hardware requirements, security patching, auditability, and whether the model can be safely used with government registry data.

Security Position

- No OpenAI API is used unless deliberately configured.
- No external model calls are made in the default setup.
- Public applicants do not receive Staff AI access.
- Nursing Council and Medical Board scope checks still apply before AI context is built.
- Model answers are guidance only; official decisions require registrar workflow and audit logs.

Official References

- Ollama API reference: <https://docs.ollama.com/api>
- Ollama OpenAI-compatible local API: <https://docs.ollama.com/openai>
- llama.cpp OpenAI-compatible server: <https://www.mintlify.com/ggml-org/llama.cpp/inference/server>